EM wave in a vacuum derivation

$\nabla\cdot\vec{E}=0 \quad \nabla\cdot\vec{H}=0 \quad \nabla\times\vec{E}=-\frac{\partial(\mu\vec{H})}{\partial t} \quad \nabla\times\vec{H}=\varepsilon_0\frac{\partial\vec{E}}{\partial t}$

$\nabla\times\nabla\times\vec{A}=\nabla(\nabla\cdot\vec{A})-\nabla^2\vec{A}$

$\Downarrow$

E:

$\nabla\times\nabla\times\vec{E}=-\nabla\times\frac{\partial(\mu\vec{H})}{\partial t}=-\mu\frac{\partial}{\partial t}(\nabla\times\vec{H})=-\varepsilon_0\mu_0\frac{\partial^2}{\partial t^2}\vec{E}$

$\nabla(\nabla\cdot\vec{E})-\nabla^2\vec{E}=-\nabla^2\vec{E} \xleftarrow{=}$

$\Rightarrow$ $\nabla^2\vec{E}=\varepsilon_0\mu_0\frac{\partial^2\vec{E}}{\partial t^2}$ in vacuum

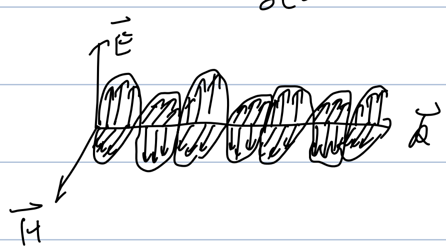
$\Rightarrow$ $v=c=\frac{1}{\sqrt{\mu_0\varepsilon_0}}$

B:

$\nabla\times\nabla\times\vec{H}=\nabla\times\left(\varepsilon_0\frac{\partial\vec{E}}{\partial t}\right)=\varepsilon_0\frac{\partial}{\partial t}\left(\nabla\times\vec{E}\right)=\varepsilon_0\mu_0\frac{\partial}{\partial t}\left(-\frac{\partial}{\partial t}\vec{H}\right)=-\varepsilon_0\mu_0\frac{\partial^2}{\partial t^2}\vec{B}$

$\nabla(\nabla\cdot\vec{H})-\nabla^2\vec{H}=-\nabla^2\vec{H} \xleftarrow{=}$

$\Rightarrow$ $\nabla^2\vec{H}=\varepsilon_0\mu_0\frac{\partial^2\vec{H}}{\partial t^2}$ in vacuum



$\nabla^2\vec{E}=\varepsilon_0\mu_0\frac{\partial^2\vec{E}}{\partial t^2}=\varepsilon_0\mu_0\omega^2\vec{E}=\beta^2\vec{E} \Rightarrow \vec{E}=E_0e^{j(\omega t-\beta z)}\hat{x}$ in vacuum

$\nabla^2\vec{H}=\varepsilon_0\mu_0\frac{\partial^2\vec{H}}{\partial t^2}=\varepsilon_0\mu_0\omega^2\vec{H}=\beta^2\vec{H} \Rightarrow \vec{H}=H_0e^{j(\omega t-\beta z)}\hat{y}$

$(\beta=\omega\sqrt{\varepsilon_0\mu_0})$

$\nabla\times\vec{H}=\begin{bmatrix}\hat{x} & \hat{y} & \hat{z}\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ H_x & \underline{H_y} & H_z\end{bmatrix}=-\frac{\partial}{\partial z}H_y\hat{x}=j\beta H_0e^{j(\omega t-\beta z)}\hat{x}$

$=\varepsilon_0\frac{\partial\vec{E}}{\partial t}=j\omega\varepsilon_0E_0e^{j(\omega t-\beta z)}\hat{x}$

$$\Rightarrow \frac{E_0}{H_0} = \frac{\beta}{\omega \varepsilon_0} = \sqrt{\frac{\mu_0}{\varepsilon_0}} = \eta_0 \approx 377\Omega$$

characteristic impedance.
frequency-independent.

How about in Matter ?

no charge

$\vec{\nabla} \cdot \vec{E} = 0 \quad \vec{\nabla} \cdot \vec{H} = 0 \quad \vec{\nabla} \times \vec{E} = -\frac{\partial(\mu \vec{H})}{\partial t} \quad \vec{\nabla} \times \vec{H} = \sigma \vec{E} + \varepsilon \frac{\partial \vec{E}}{\partial t}$

From $\vec{\nabla} \times \vec{\nabla} \times \vec{A} = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A}$

For $\vec{E}$:

$LHS = \vec{\nabla} \times \left(-\frac{\partial}{\partial t}(\mu \vec{H})\right) = -\mu \frac{\partial}{\partial t}(\vec{\nabla} \times \vec{H}) = -\mu \frac{\partial}{\partial t}\left(\sigma \vec{E} + \varepsilon \frac{\partial \vec{E}}{\partial t}\right)$

$= -\mu\sigma \frac{\partial}{\partial t} \vec{E} - \mu\varepsilon \frac{\partial^2}{\partial t^2} \vec{E}$

$RHS = -\nabla^2 \vec{E}$

$\Rightarrow \nabla^2 \vec{E} = \mu\sigma \frac{\partial}{\partial t} \vec{E} + \mu\varepsilon \frac{\partial^2}{\partial t^2} \vec{E}$

$= (j\omega\mu\sigma - \omega^2 \mu\varepsilon) \vec{E}$

$= \gamma^2 \vec{E}$

$\gamma = \alpha + j\beta = \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)}$

$\Rightarrow \vec{E} = E_0 \cdot e^{j\omega t} \cdot e^{-\gamma z} \, \hat{x} = E_0 \cdot e^{j(\omega t - \beta z)} \cdot e^{-\frac{z}{\delta}} \, \hat{x}$

Similarly, $\vec{H} = H_0 \cdot e^{j\omega t} \cdot e^{-\gamma z} \, \hat{y} = H_0 \cdot e^{j(\omega t - \beta z)} \cdot e^{-\frac{z}{\delta}} \, \hat{y}$

$\vec{\nabla} \times \vec{E} = -\frac{\partial(\mu H)}{\partial t}$

$$\begin{cases} LHS = \frac{\partial}{\partial z} E_x \, \hat{y} = -\gamma E_0 \cdot e^{j\omega t} \cdot e^{-\gamma z} \cdot \hat{y} \\ RHS = -j\omega\mu \cdot H_0 \cdot e^{j\omega t} \cdot e^{-\gamma z} \end{cases}$$

$\Rightarrow \eta = \frac{E_0}{H_0} = \frac{j\omega\mu}{\gamma} = \sqrt{\frac{j\omega\mu}{(\sigma + j\omega\varepsilon)}}$